

Hardware specification

Component blocks

- Type / organization
- Dimensions / capacity
- Bandwidth / clock
- Access cycles / energy
- Precision / format

select

Optional module blocks

Weight decompressor
SFU / KV-cache unit

Component library

PE

MAC lanes + I / W / O local buffers

Array / interconnect

Spatial / systolic / tree; bus / mesh

Global buffer

Shared / partitioned; double / bypass

Package network

Chip grid + multicast delivery

DRAM

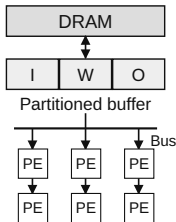
Analytical model / DRAMsim3

Optimization modules

Instantiate & connect at initialization

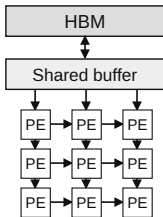
(a) Spatial

Eyeriss-like



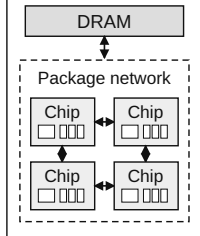
(b) Systolic

TPU-like



(c) Multi-chip

Simba-like



I / W / O: input / weight / output SFU: special function unit